

Roll No.

MID SEMESTER EXAM MARCH-2025

Name of the Program: B.E.(E&C)

Semester : VIII

Name of the Course: Neural Network and Machine Learning

Course Code : TEC 856

Time : 90 minutes

Maximum Marks : 50 Marks

Note :

- I. Answer all the questions by choosing one of the sub questions.
- II. Each Question carries 10 Marks.

Q1	(10 Marks)	CO1																								
(a)	Explain the structure of a biological neuron and illustrate the model used for neural computing																									
	OR																									
(b)	Explain McCulloch Pitts model of neural network and illustrate implementation of OR operation																									
Q2	(10 Marks)	CO1																								
(a)	Design a pattern classifier using basic neural network model with S as a set of patterns to be classified in to classes A and B S={000, 001, 010,011, 100, 101, 110, 111} A ={000, 001, 010, 100} B={011,101,110,111}																									
	OR																									
(b)	Illustrate implementation using neural network of the following table (a) Truth Table <table><tr><th>x_1</th><th>x_2</th><th>x_1 AND (NOT x_2)</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	x_1	x_2	x_1 AND (NOT x_2)	0	0	0	0	1	0	1	0	1	1	1	0										
x_1	x_2	x_1 AND (NOT x_2)																								
0	0	0																								
0	1	0																								
1	0	1																								
1	1	0																								
Q3	(10 Marks)	CO1, CO2																								
(a)	Explain the structure of Perceptron. Implement AND Operation using the Perceptron																									
	OR																									
(b)	Explain the XOR Problem in Neural Network and suggest a solution																									
Q4	(10 Marks)	CO1,CO2																								
(a)	What are the different types of learning schemes for neural network. Explain with examples																									
	OR																									
(b)	With Hebb Learning Rule train the neural network to realise the following table Training set <table><tr><th colspan="3">Input Patterns</th><th>Output</th></tr><tr><th>x_0</th><th>x_1</th><th>x_2</th><th>t</th></tr><tr><td>+1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>+1</td><td>1</td><td>-1</td><td>-1</td></tr><tr><td>+1</td><td>-1</td><td>1</td><td>-1</td></tr><tr><td>+1</td><td>-1</td><td>-1</td><td>-1</td></tr></table>	Input Patterns			Output	x_0	x_1	x_2	t	+1	1	1	1	+1	1	-1	-1	+1	-1	1	-1	+1	-1	-1	-1	
Input Patterns			Output																							
x_0	x_1	x_2	t																							
+1	1	1	1																							
+1	1	-1	-1																							
+1	-1	1	-1																							
+1	-1	-1	-1																							
Q5	(10 Marks)	CO2																								
(a)	With example explain Perceptron Learning Rule																									
	OR																									
(b)	Explain any two activation functions																									

